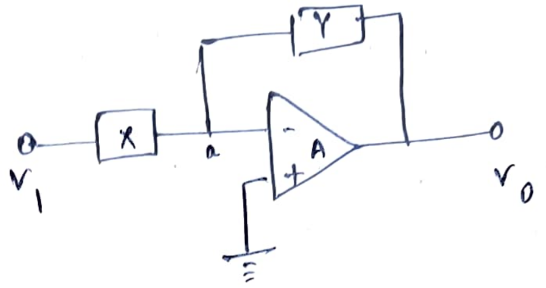
(1)

Given $X = L = (SL \text{ in } \Omega)$

$Y = R$



$\Rightarrow$ Applying virtual ground concept $\Rightarrow$ $v_a = 0$

Nodal at point a

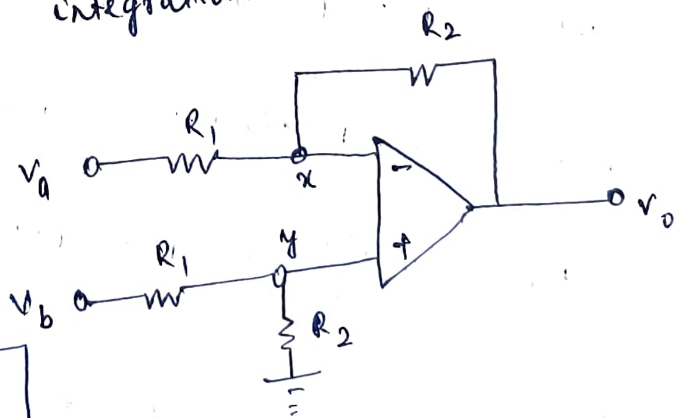
$$\frac{v_a - v_1}{SL} + \frac{v_a - v_0}{R} = 0$$

$$\Rightarrow -\frac{v_1}{SL} - \frac{v_0}{R} = 0$$

$$\Rightarrow \boxed{\frac{v_0}{v_1} = -\frac{R}{SL}} \quad \text{(In frequency domain)}$$

$$\Rightarrow \boxed{v_0 = -\frac{R}{L}\int v_1\,dt} \quad \text{(in time domain)}$$

$\Rightarrow$ The ckt performs integration operation.



(2) Since opamp is
(a) ideal so, $v_x = v_y$

$$\boxed{v_y = v_b \frac{R_2}{R_1 + R_2} = v_x}$$

Now, applying nodal analysis at x

$$\frac{V_x - V_a}{R_1} + \frac{V_x - V_o}{R_2} = 0$$

Put $V_x = V_b \frac{R_2}{R_2 + R_1}$

we get, $\boxed{V_o = \frac{R_2}{R_1}(V_b - V_a)}$ — (A)

(b) voltage subtraction

(c) $R_1 = R_2 = 50\ k\Omega$, $V_a = -3 + \sin\omega t$, $V_b = \sin\omega t + 2\sin\omega t$

so, (A) $\Rightarrow$ $\boxed{V_o = 3 + 2\sin\omega t}$

(3) Given $SR = 0.5\ V/\mu s$, $f = 3000\ Hz$, max^m o/p $= 12V$ (Peak)

Let $V_o = 12\sin\omega t$ where $\omega = 2\pi \times 3000 = 6000\pi$ rad/s

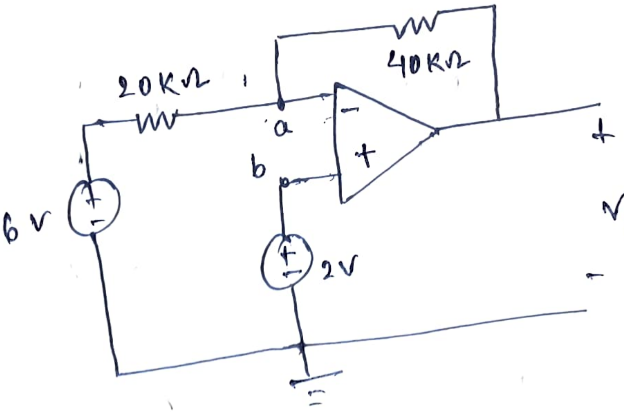
$$S.R. = \frac{dV_o}{dt} = 12\omega\cos\omega t$$

$$\left|\frac{dV_o}{dt}\right| = 12 \times 6000\pi = 0.226\ V/\mu s$$

as calculated S.R. is less than given S.R. ($0.5\ V/\mu s$)

so, opamp can handle its o/p without slew. slew will not happen.

(4)



Since the opamp is ideal,

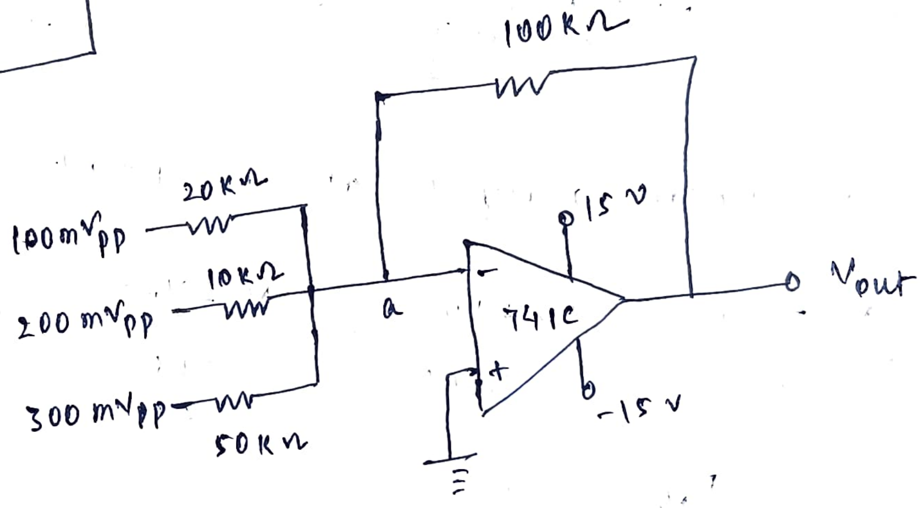
$v_b = v_a = 2V$

apply nodal analysis at a

$$\frac{v_a - 6}{20\,k\Omega} + \frac{v_a - v_o}{40\,k\Omega} = 0$$

$\Rightarrow$ $\boxed{v_o = -6V}$

(5)



$v_a = 0$ (virtual ground)

apply nodal analysis at v_a

we get $V_{out} = -5 \times 0.1 V_{pp} - 10 \times 0.2 V_{pp} - 2 \times 0.3 V_{pp}$

$V_{out} = -3.1 V_{pp}$

so, $\boxed{V_{out} = 3.1\,V \text{ (Peak to Peak)}}$

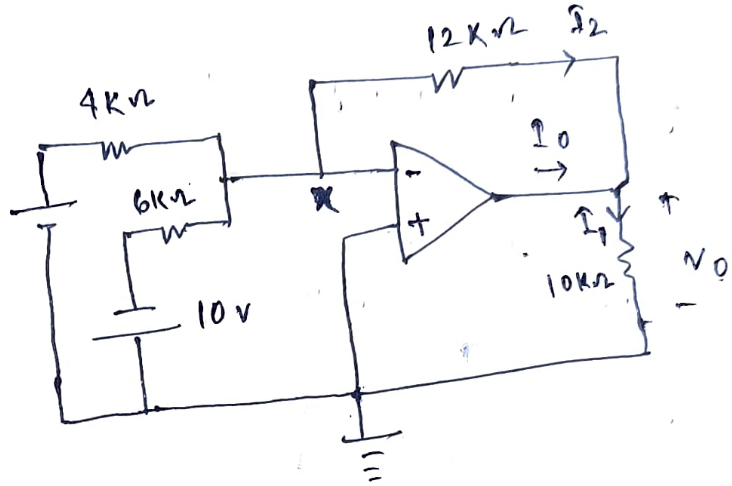
since $V_{out} < V_{saturation}$

(b) Slew Rate = $\frac{\text{change in o/p voltage}}{\text{time duration}}$

$$S.R. = \frac{0.25}{0.1} \; V/\mu s$$

$$\boxed{S.R. = 2.5 \; V/\mu s}$$

(7)



(a) Given $V_a = 4V$

$V_x = 0$ [virtual ground concept]

apply nodal analysis at x point

$$\frac{V_x - V_a}{4000} + \frac{V_x + 10}{6000} + \frac{V_x - V_o}{12000} = 0$$

$$\boxed{V_o = 8V}$$

Now, $I_o = I_1 - I_2$ —— (A)

$$I_1 = \frac{V_o}{10 K\Omega} = 0.8 \; mA$$

$$I_2 = \frac{-8}{12 K\Omega} = -0.67 \; mA$$

So, (A) ⇒ $\boxed{I_o = 1.47 \; mA}$

(b) once again, writing nodal analysis equation at x ($V_x = 0$)

$$-\frac{V_a}{4000} + \frac{10}{6000} = \frac{V_o}{12000}$$

$$\boxed{V_o = -3V_a + 20} \quad \text{— (1)}$$

To operate in linear region, $V_o < V_{saturation}$

i.e. $-12 < V_o < 12$ (volt)

<u>case 1</u>

when $V_o > 12$

(1) $\Rightarrow$ $-3V_a + 20 > 12$

$$\boxed{V_a < \frac{32}{3} \text{ V}}$$

<u>case 2</u> when $V_o < 12$

(1) $\Rightarrow$ $-3V_a + 20 < 12$

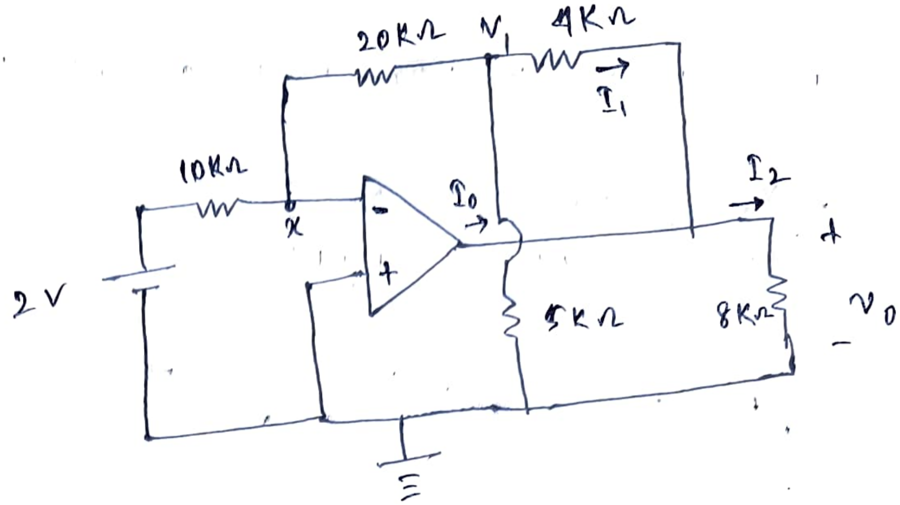
$$\boxed{V_a > 8/3 \text{ V}}$$

so, from case 1 & case 2

$$\boxed{\frac{8}{3} < V_a < \frac{32}{3} \text{ (volt)}}$$

$\Rightarrow$ range of V_a for linear operation

(8)



$V_x = 0$ (virtual ground)

writing nodal equations at node x

$$\frac{V_x - 2}{10000} + \frac{V_x - V_1}{20000} = 0$$

$$\boxed{V_1 = -4\ V}$$

writing nodal equation at node V_1

$$\frac{V_1}{5000} + \frac{V_1 - V_x}{20000} + \frac{V_1 - V_0}{4000} = 0$$

$$\boxed{V_0 = -8\ V}$$

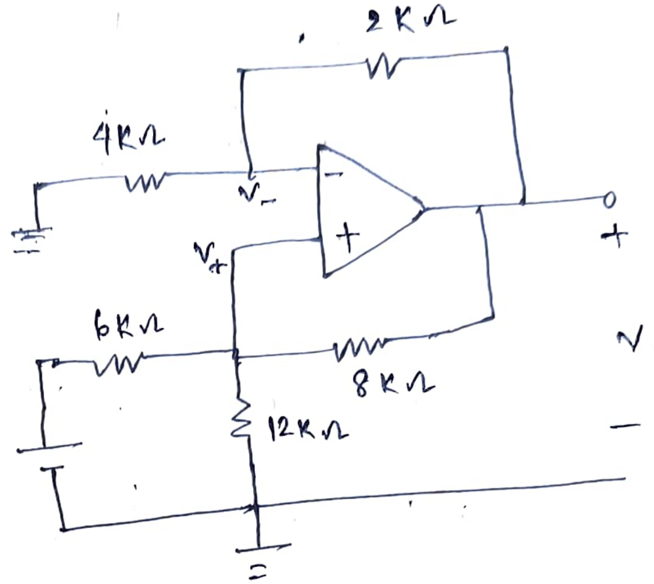
so, $I_0 = I_2 - I_1$ —— (A)

$$I_2 = \frac{V_0}{8K\Omega} = -1\ mA$$

(A) ⇒ $\boxed{I_0 = -2\ mA}$

$$I_1 = \frac{V_1 - V_0}{4K\Omega} = 1\ mA$$

(9)



writing nodal analysis equation at v_+

$$\frac{v_+}{12000} + \frac{v_+ - 6}{6000} + \frac{v_x - v_o}{8000} = 0 \quad \text{—— (1)}$$

Also, $v_+ = v_-$ [virtual ground]

writing nodal eqn at v_-

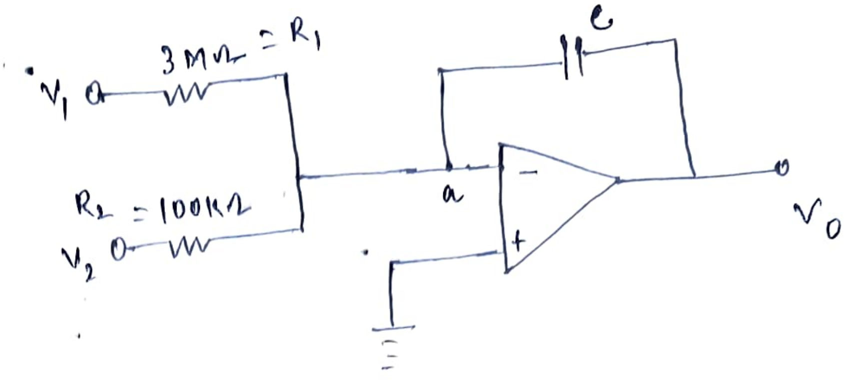
$$\frac{v_- - 0}{4000} + \frac{v_- - v_o}{2000} = 0 \quad \text{—— (2)}$$

solve eqn (1) & (2)

$$v^- = v^+ = \frac{2v_o}{3}$$

& $\boxed{v_o = 8\ V}$

(10)



$R_1 = 3\,M\Omega$, $R_2 = 100\,k\Omega$, $C = 2\,\mu F$

$v_1 = 100\cos 2t$ mV, $v_2 = 0.5\,t$ mV

Now,

$v_a = 0$ [virtual ground]

writing nodal eqn at node a in s - domain

$$-\frac{V_1}{R_1} - \frac{V_2}{R_2} - V_0 \cdot sC = 0$$

$$\Rightarrow V_0 = -\frac{1}{R_1 R_2 C s}\left[V_1 R_2 + V_2 R_1\right]$$

in time domain

$$\Rightarrow v_0 = -\frac{1}{R_1 R_2 C}\left[\int_0^t v_1 R_2\,dt + \int_0^t v_2 R_1\,dt\right]$$

Putting all values ....., we get

$$v_0 = -\frac{1}{R_1 R_2 C}\left[\frac{10^{-2}}{2} R_2 \sin 2t + 0.25 \times 10^{-3} R_1 t^2\right]$$

$$\boxed{v_0 = -(83.3 \sin 2t + 1.25\,t^2)\ \text{mV}}$$